

Ava Hope Cheung

(516) 643-9325 | avacheung17@gmail.com | [linkedin.com/in/avacheung](https://www.linkedin.com/in/avacheung) | github.com/avacheungx | <https://avacheungx.github.io/Juno-Website/>

EDUCATION

New York University | Courant Institute of Mathematical Sciences - New York, NY

September 2022 - May 2026

B.A. in Computer Science | Minor in Web Programming and Applications

Relevant Courses: Data Structures, Computer Architectures, Basic Algorithms, Operating Systems, Software Engineering, Agile Development and DevOps, Artificial Intelligence, Computer Networks

TECHNICAL EXPERIENCE

NYU Stern School of Business | New York, NY

IT Support Lead - Classroom Technology

January 2024 - Present

- Led Tier 1 & Tier 2 incident response, diagnosing hardware, software, and network connectivity issues across classroom IT systems, reducing operational downtime by 20%.
- Supported classroom AV systems, instructor workstations, and network-connected devices across multiple lecture halls to ensure reliable academic operations.
- Managed IT asset inventory and device lifecycle tracking, maintaining compliance with technology and security policies.
- Documented troubleshooting workflows and trained new IT staff on support procedures, AV systems, and classroom technology infrastructure.

IT Help Desk Analyst

August 2023 - Present

- Managed incoming IT support tickets through ServiceNow, Genesys, and ServiceLink, resolving hardware, software, and connectivity issues for faculty and staff.
- Diagnosed recurring technical issues and conducted root cause analysis, recommending workflow improvements that streamlined support procedures.
- Performed system checks and infrastructure audits to ensure classroom and office technology remained operational.

PROJECTS

SyncWave - Music Management Web App | React, Node.js, MongoDB

November 2025

- Built a music discovery and playlist management web app for DJs using React, Node.js, and MongoDB.
- Integrated Spotify Web API and YouTube Data API v3 for cross-platform track search with automatic metadata extraction (BPM, key, genre) and token management.
- Built a real-time track compatibility tool with visual matching indicators to assist in DJ set planning.
- Implemented RESTful APIs with session-based authentication, advanced filtering, and Tailwind-based UI.

WeensyOS - Virtual Memory and Process Isolation | C, x86-64, QEMU, gdb

April 2025

- Built kernel memory isolation and per-process virtual address spaces in a teaching operating system by modifying page table mappings and access permissions.
- Added virtual page allocation and overlapping address spaces, enabling processes to map different physical pages at the same virtual addresses.
- Implemented the fork() system call by copying page tables and duplicating writable user pages.
- Debugged low-level kernel and memory faults using QEMU, gdb, assertions, and log-based tracing.

Resume Analyzer - Microservices Resume Evaluation Tool | FastAPI, Flask, Docker

April 2025

- Led a team of four engineers building a resume analysis platform using FastAPI, Flask, MongoDB, and Docker.
- Built CI/CD pipelines with GitHub Actions, achieving 80%+ code coverage and reducing deployment time by 70%.
- Coordinated Agile sprint planning, standups, and code reviews.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, JavaScript, TypeScript, Bash

Frameworks & Libraries: React.js, FastAPI, Flask, Express.js, Tailwind CSS

Tools & Platforms: Docker, Git, MongoDB, PostgreSQL, SQLite, Figma, VS Code

IT Systems & Infrastructure: Windows & macOS troubleshooting, endpoint support, hardware diagnostics, network troubleshooting, AV system support, IT asset management

IT Service Management: ServiceNow, Genesys, ServiceLink, Jira, incident management, ticket triage, root cause analysis, technical documentation

CERTIFICATIONS

CompTIA A+: In Progress (Expected May 2026)